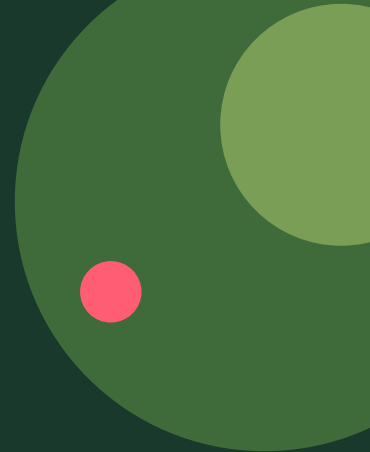




PROJECT 1 / SIXTY SECONDS + BLOOM

# BLOOM

Game Design Document

| FORMAT             | GENRE                     | PLATFORM            |
|--------------------|---------------------------|---------------------|
| Online multiplayer | 2D top-down battle royale | Desktop web browser |

A survival shootout in a procedurally generated, overgrown garden where the safe zone contracts in sixty-second breaths and every carried seed becomes more valuable with time.

SECTION 01

# Core Loop

Explore the generated garden, collect and manage a rifle, ammunition, and one growing flower, fight other players, and reposition inside each randomized safe zone until only one player remains.

The repeating decision is whether to keep moving for weapons and ammunition, engage an opponent, or protect a carried seed long enough for it to become a stronger heal.

SECTION 02

# The Sixty-Second Shape

Sixty seconds is the duration of each active storm contraction, not a visible match countdown. The storm then holds still for 10 seconds before another randomized contraction begins. The sequence repeats until one survivor remains.

| 0:00                                                                                                                                                                        | 0:30                                                                                                                                                                                                                    | 0:55                                                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Players enter unarmed. They spread through a new procedural garden, auto-collect rifles, ammunition, and dormant seeds, and react to the first randomized safe-zone target. | The storm is halfway through its contraction. Carried seeds are gaining healing value, cover routes are narrowing, and armed players must choose between pursuit, repositioning, and preserving a stronger future heal. | The safe space is nearly at its target size. Players are compressed into decisive encounters, mature flowers can swing a fight, and the approaching 10-second calm offers a brief reset if more than one survivor remains. |

If the first contraction does not produce a winner, the same pressure arc returns with a smaller, newly positioned safe zone. The fourth play changes because the map, loot, seeds, player routes, and future safe zones are regenerated.

SECTION 03

# Reading of Bloom

Bloom is a resource that matures only after a player commits to carrying it. Seeds spawn dormant at random positions. Walking over one places it in the player's single flower slot and begins its growth. Its stored healing starts at 1 HP, rises by 1 HP every 2 seconds, and stops at 50 HP. Pressing E consumes it, restores its current value without exceeding the player's 100 HP maximum, and removes it for the rest of the match.

| SEED                    | BUDDING                  | BLOOMED                               |
|-------------------------|--------------------------|---------------------------------------|
| 1-16 stored HP          | 17-33 stored HP          | 34-50 stored HP                       |
| Weak now, strong later. | A useful emergency heal. | Maximum visual and mechanical payoff. |

The theme lives in a combat decision: consume a small heal now, or risk losing the match while waiting for the same seed to bloom into a larger one. Flowers are the only healing source, so growth is never decorative.

# Art Direction

|                       |                                                                                                                                                                                  |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| World in one sentence | A procedurally assembled, overgrown garden of dense plants, broken stone paths, hedges, rocks, and abandoned garden structures.                                                  |
| Camera and framing    | Strictly top-down 2D with no angle or simulated depth. The camera follows the active player with smooth interpolation and transfers to the current target during spectator mode. |
| Light                 | Flat, even lighting across all play space. Readability comes from color and silhouette rather than shadows, highlights, or atmospheric effects.                                  |
| Edge and detail       | Clean flat shapes, crisp silhouettes, consistent outlines, and limited interior detail. Objects must remain recognizable at gameplay scale.                                      |
| Three words of mood   | Lush. Tense. Untamed.                                                                                                                                                            |
| Style suffix          | clean flat-shape game sprite, strict top-down 2D, crisp silhouette, limited detail, even color, no depth                                                                         |
| Avoid list            | No gradients, textures, perspective, shadows, glow effects, bloom rendering, or realistic rendering.                                                                             |
| Anchor asset          | The fully bloomed flower sprite. It establishes the clean silhouette, controlled detail, and bloom-pink accent used to signal the game's most important resource.                |

PALETTE / FIVE VALUES, ONE ACCENT

|             |            |         |                 |                     |
|-------------|------------|---------|-----------------|---------------------|
| #18392B     | #3F6B3A    | #7B9E57 | #A7A58E         | #FF5D73             |
| Deep forest | Leaf green | Moss    | Weathered stone | Bloom pink - accent |

Bloom pink is reserved for flowers and other highest-priority cues. The remaining four colors establish terrain, foliage, cover, interface neutrals, and player readability.

# Scope

IN

- Browser-based HTML, CSS, and JavaScript client, with the public build hosted on GitHub Pages.
  - AWS-hosted real-time multiplayer server with private rooms, a lobby, and a host-controlled Start button.
  - No fixed design cap on room population; generated map size, obstacle count, loot, and spawn spacing scale with lobby population, subject to tested server capacity.
  - A new procedural overgrown-garden map each match, with validated player spawns and randomized rocks, walls, rifles, ammunition, dormant seeds, and safe-zone targets.
  - Players spawn unarmed with 100 HP. Pickups are collected automatically on contact.
- One automatic rifle: 30-round full magazine on pickup, 5 damage per hit, plus ammunition pickups. Fire rate and bullet spread are developer-tunable constants.
  - WASD movement, mouse aiming, hold left-click to fire, R to reload, and E to consume the carried flower.
  - Storm contracts continuously for 60 seconds, pauses for 10 seconds, and repeats toward randomized nested safe zones. Outside players lose 1 HP per second.
  - Permanent elimination at 0 HP. Spectators follow their killer; if that player dies, the camera transfers along the killer chain. Late joiners spectate a random living player until the next match.
  - One carried flower slot, three visual growth sprites, and no flower regrowth after collection or consumption.

CUT BEFORE PRODUCTION

|    |                            |                                                                                                                 |
|----|----------------------------|-----------------------------------------------------------------------------------------------------------------|
| 01 | Multiple weapon types      | Only the rifle ships in this version; weapon variety would multiply balancing, art, audio, and networking work. |
| 02 | Conventional healing items | No medkits or bandages. Flowers remain the single healing system and the clearest expression of Bloom.          |
| 03 | Classes and equipment      | No character classes, special abilities, armor, attachments, or equipment progression.                          |

# What I Expect to Be Fragile

The most fragile part is the interaction between real-time networking, procedural generation, and a continuously changing safe zone. Every client must agree about player positions, bullets, pickups, flower timers, obstacle collisions, health, elimination, spectator targets, and storm geometry. The AWS server therefore needs to be authoritative for match state, while clients render and predict only enough to feel responsive.

| SYSTEM                             | EXPECTED FAILURE                                                                                                                  | PLANNED RESPONSE                                                                                                                                                                                                     |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Network synchronization            | Latency or disagreement can make hits, pickups, storm damage, and deaths appear inconsistent.                                     | Keep combat and health server-authoritative; test with multiple browsers and artificial latency early.                                                                                                               |
| Procedural map validity            | Random obstacles can trap players, block required routes, overlap loot, or produce unfair opening spawns.                         | Generate from one server seed, validate clear spawn radii and connected walkable space, then regenerate invalid layouts.                                                                                             |
| Scaling without a fixed player cap | Population can exceed the map, network, collision, or spectating capacity that has actually been tested.                          | Scale map and loot by population, expose tuning constants, monitor server load, and document the tested practical capacity.                                                                                          |
| Sixty-second fit                   | Because the match ends with one survivor rather than at 1:00, some rounds may run longer than the assignment's target experience. | Use the first 60-second contraction to force early contact, record playtest match lengths, and tune map scale, storm endpoint, and loot density until informed players reach decisive outcomes near the first cycle. |
| Spectator chain and disconnects    | The watched killer may die or leave while late joiners and eliminated players are observing different targets.                    | Store a server-owned spectator target and transfer it atomically to the killer or another living player.                                                                                                             |

PLAYTEST  
QUESTION

Does waiting for a flower to bloom create a meaningful survival decision while the storm is forcing movement, or does the optimal choice become obvious?